

Power BI Assignment 1 – Data Transformation and modeling

Import Data

- List of orders.csv file is imported in to the PowerBI using 'Get data' in home tab. Select text/csv. Click on load after selecting the csv file.

List of Orders.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on entire dataset

Column1	Column2	Column3	Column4	Column5
Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur

Extract Table Using Examples | Load | Transform Data | Cancel

- Using 'Transform Data' option, it is opened in power query

Power Query Editor interface showing the 'List of Orders' query loaded from the CSV file. The data is displayed in a table with columns: Order ID, Order Date, CustomerName, State, and City. The 'Query Settings' pane on the right shows the 'Source' step and the 'Change Type' option under 'APPLIED STEPS'.

Table Data:

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur

- By using the 'new source' option in the home tab, 'Order details.csv' and 'Sales target.csv' are imported to the power query editor.

Order Details.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on entire dataset

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1275	-1148	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankerchief
B-25601	80	-56	4	Electronics	Electronic Games
B-25602	168	-111	2	Electronics	Phones
B-25602	424	-272	5	Electronics	Phones
B-25602	2617	1151	4	Electronics	Phones
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	1355	-60	5	Clothing	Trousers
B-25603	24	-30	1	Furniture	Chairs
B-25603	193	-166	3	Clothing	Saree
B-25603	180	5	3	Clothing	Trousers
B-25603	116	16	4	Clothing	Stole
B-25603	107	36	6	Clothing	Stole
B-25603	12	1	2	Clothing	Hankerchief
B-25603	38	18	1	Clothing	Kurti
B-25604	65	17	2	Clothing	T-shirt
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree
B-25606	87	4	2	Clothing	Shirt
B-25607	50	15	4	Clothing	Leggings

Extract Table Using Examples

OK

Cancel

Sales target.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on entire dataset

Month of Order Date	Category	Target
4/18/2026	Furniture	10400
5/18/2026	Furniture	10500
6/18/2026	Furniture	10600
7/18/2026	Furniture	10800
8/18/2026	Furniture	10900
9/18/2026	Furniture	11000
10/18/2026	Furniture	11100
11/18/2026	Furniture	11300
12/18/2026	Furniture	11400
1/19/2026	Furniture	11500
2/19/2026	Furniture	11600
3/19/2026	Furniture	11800
4/18/2026	Clothing	12000
5/18/2026	Clothing	12000
6/18/2026	Clothing	12000
7/18/2026	Clothing	14000
8/18/2026	Clothing	14000
9/18/2026	Clothing	14000
10/18/2026	Clothing	16000
11/18/2026	Clothing	16000
12/18/2026	Clothing	16000
1/19/2026	Clothing	16000

Extract Table Using Examples

OK

Cancel

Data Transformation

- To restrict number of rows to first 500 rows in 'List of orders', Select the list of orders data, go to 'home' tab – 'Keep Rows' – on the drop down menu select 'keep top rows' – a dialog box will appear on it type '500'.
- To change the data type, go the icon next to the filter arrow in table headers – click on the icon and from the drop down menu-Select the data type you want. For order date, 'Date' type is selected
- For Amount and target, Data type is converted into 'Decimal numbers'.
- To ensure capitalization in each word in 'Customer Name' Column, Select the column – Go to 'transform' tab – Select 'Format' option in 'text column' – on the drop down menu, Select 'Capitalize Each Word'. Now each word is capitalized.
- Use Ctrl key and select which of the columns you want to merge, and select 'merge columns' option in 'add column' tab. Give a title for the column and press ok.

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Query Settings

Layout Data Preview Columns Parameters Advanced Dependencies

Queries [3]

Sales target

Order Details

Orders Data

Order Date CustomerName State City Location

1 4/1/2018 Bharat Gujarat Ahmedabad Gujarat,Ahmedabad

2 4/1/2018 Pearl Maharashtra Pune Maharashtra,Pune

3 4/3/2018 Jahan Madhya Pradesh Bhopal Madhya Pradesh,Bhopal

4 4/3/2018 Divsha Rajasthan Jaipur Rajasthan,Jaipur

5 4/5/2018 Kasheen West Bengal Kolkata West Bengal,Kolkata

6 4/6/2018 Hazel Karnataka Bangalore Karnataka,Bangalore

7 4/6/2018 Sonakshi Jammu and Kashmir Jammu and Kashmir,Kashmir

8 4/8/2018 Aarushi Tamil Nadu Chennai Tamil Nadu,Chennai

9 4/9/2018 Jitesh Uttar Pradesh Lucknow Uttar Pradesh,Lucknow

10 4/9/2018 Yogesh Bihar Patna Bihar,Patna

11 4/11/2018 Anita Kerala Thiruvananthapuram Kerala,Thiruvananthapuram

12 4/12/2018 Shrichand Punjab Chandigarh Punjab,Chandigarh

13 4/12/2018 Mukesh Haryana Chandigarh Haryana,Chandigarh

14 4/13/2018 Vandana Himachal Pradesh Simla Himachal Pradesh,Simla

15 4/15/2018 Bhavna Sikkim Gangtok Sikkim,Gangtok

16 4/15/2018 Kanak Goa Goa Goa,Goa

17 4/17/2018 Sagar Nagaland Kohima Nagaland,Kohima

18 4/18/2018 Manju Andhra Pradesh Hyderabad Andhra Pradesh,Hyderabad

19 4/18/2018 Ramesh Gujarat Ahmedabad Gujarat,Ahmedabad

20 4/20/2018 Sarita Maharashtra Pune Maharashtra,Pune

21 4/20/2018 Deepak Madhya Pradesh Bhopal Madhya Pradesh,Bhopal

22 4/22/2018 Monisha Rajasthan Jaipur Rajasthan,Jaipur

23 4/22/2018 Arhanu West Bengal Kolkata West Bengal,Kolkata

Query Settings

PROPERTIES

Name

Orders Data

All Properties

APPLIED STEPS

Source

Change Type

Promoted Headers

Changed Type

Kept First Rows

Changed Type with Locale

Capitalized Each Word

Inserted Merged Column

6 COLUMNS, 500 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:53 PM

- To create the profit margin column, First go to 'add column' Tab, Select 'Custom column' and on the dialog box write the necessary equations and press ok.
The data type of the profit margin is changed into percentage.

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula ⓘ

Available columns

- Order ID
- Amount
- Profit
- Quantity
- Category
- Sub-Category

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples Column Function Duplicate Column

General

Conditional Column Index Column Merge Columns Format Parse From Text

Statistics Standard Scientific From Number

Trigonometry Rounding Information

Date Time Duration From Date & Time

Queries [3]

Sales target

Order Details

List of Orders

fx = Table.TransformColumnTypes(#"Added Custom",{{"Profit Margin", Percentage.Type}})

	1 ² Profit	1 ² Quantity	1 ² Category	1 ² Sub-Category	% Profit Margin
1	1275	-1148	7 Furniture	Bookcases	-90.04%
2	66	-12	5 Clothing	Stole	-18.18%
3	8	-2	3 Clothing	Hankkerchief	-25.00%
4	80	-56	4 Electronics	Electronic Games	-70.00%
5	168	-111	2 Electronics	Phones	-66.07%
6	424	-272	5 Electronics	Phones	-64.15%
7	2617	1151	4 Electronics	Phones	43.98%
8	561	212	3 Clothing	Saree	37.79%
9	119	-5	8 Clothing	Saree	-4.20%
10	1355	-60	5 Clothing	Trousers	-4.43%
11	24	-30	1 Furniture	Chairs	-125.00%
12	193	-166	3 Clothing	Saree	-86.01%
13	180	5	3 Clothing	Trousers	2.78%
14	116	16	4 Clothing	Stole	13.79%
15	107	36	6 Clothing	Stole	33.64%
16	12	1	2 Clothing	Hankkerchief	8.33%
17	38	18	1 Clothing	Kurti	47.37%
18	65	17	2 Clothing	T-shirt	26.15%
19	157	5	9 Clothing	Saree	3.18%
20	75	0	7 Clothing	Saree	0.00%
21	87	4	2 Clothing	Shirt	4.60%
22	50	15	4 Clothing	Leggings	30.00%
23	1964	-1864	5 Furniture	Tables	-196.66%

7 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:19 AM

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Use First Row as Headers

Change Type

Added Custom

Changed Type

- For creating the conditional column, Go to 'Add Column' Tab, Select conditional column, Add the conditions in the dialog box, press ok.

Power Query Editor interface showing a table with 7 columns and 499 rows. The table contains order data with columns: Order ID, Order Date, CustomerName, State, City, and Location. The data is displayed in a grid view.

Queries [3]: Sales target, Order Details, Orders Data.

Query Settings: Properties (Name: Orders Data), Applied Steps (Source, Change Type, Kept First Rows, Promoted Headers, Changed Type, Changed Type with Locale, Capitalized Each Word, Inserted Merged Column, Renamed Columns, Merged Queries).

7 COLUMNS, 499 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 11:39 AM

Handling Missing and Duplicate Data

- Missing data can be identified by many methods. Here I checked the column statistics using 'view' tab – by ticking 'column quality' and 'column profile' giving me a description of all the rows in the column.

PowerBI Assignment 1 - Data Transformation and Modeling interface showing a table with 14 columns and 999+ rows. The table contains order data with columns: Order ID, Order Date, CustomerName, State, City, and Location. The data is displayed in a grid view.

Queries [3]: Sales target, Order Details, Orders Data.

Query Settings: Properties (Name: Orders Data), Applied Steps (Source, Change Type, Promoted Headers, Changed Type, Kept First Rows, Changed Type with Locale, Capitalized Each Word, Inserted Merged Column, Merged Queries, Expanded Order Details).

Column statistics:

Column	Count	Error	Empty	Distinct	Unique	Empty string	Min	Max
B-25601	1000	0	0	343	165	0	B-25601	B-25943

Value distribution:

Column	Count
B-25656	1000
B-25757	1000
B-25851	1000
B-25651	1000
B-25751	1000
B-25653	1000
B-25850	1000
B-25855	1000
B-25655	1000
B-25753	1000
B-25650	1000

14 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 3:01 PM

No empty values were identified. If there was empty values, I would use COALESCE function to do imputation.

- You can check for duplicates using the 'column profile' and 'column quality' in 'view' tab. In order to remove duplicates, Select the whole table - go to 'home' tab – 'Remove Rows' – in the dropdown menu select 'remove duplicates'.

Sorting and Filtering Data

- In order to filter the order data's 'order date' in ascending order, select the arrow button in the 'Order Date' column header and select 'Sort Ascending' in the drop down menu.

The screenshot shows the PowerBI Desktop interface with the 'Orders Data' table selected. The 'Query Settings' pane on the right displays the 'APPLIED STEPS' section, which includes 'Sorted Rows'. The table data is sorted by 'Order Date' in ascending order. The 'Query Settings' pane also shows the 'PROPERTIES' section with 'Orders Data' as the name.

Order ID	CustomerName	State	City	Location
1	Bharat	Gujarat	Ahmedabad	Gujarat,Ahm
2	Pearl	Maharashtra	Pune	Maharashtra
3	Pearl	Maharashtra	Pune	Maharashtra
4	Bharat	Gujarat	Ahmedabad	Gujarat,Ahm
5	Pearl	Maharashtra	Pune	Maharashtra
6	Bharat	Gujarat	Ahmedabad	Gujarat,Ahm
7	Pearl	Maharashtra	Pune	Maharashtra
8	Bharat	Gujarat	Ahmedabad	Gujarat,Ahm
9	Pearl	Maharashtra	Pune	Maharashtra
10	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
11	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
12	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
13	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
14	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
15	Divsha	Rajasthan	Jaipur	Rajasthan,Jai
16	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
17	Jahan	Madhya Pradesh	Bhopal	Madhya Prac
18	Divsha	Rajasthan	Jaipur	Rajasthan,Jai
19	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
20	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
21	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
22	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
23	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
24	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
25	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
26	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
27	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
28	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
29	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C
30	Aarushi	Tamil Nadu	Chennai	Tamil Nadu,C

- To filter the focus on a specific state, Go to the column header of state and click on the down arrow button, on the search bar, search 'Tamil Nadu', press ok. All the data on Tamil Nadu will be displayed

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Query Settings

Layout Data Preview Columns Parameters Advanced Query Dependencies

Queries [3]

Sales target

Order Details

Orders Data

Table.SelectRows(#"Sorted Rows", each ([State] = "Tamil Nadu"))

Order ID	Order Date	CustomerName	State	City	Location
1	4/8/2018			Chennai	Tamil Nadu, Chennai
2	4/8/2018			Chennai	Tamil Nadu, Chennai
3	4/8/2018			Chennai	Tamil Nadu, Chennai
4	4/8/2018			Chennai	Tamil Nadu, Chennai
5	6/23/2018			Chennai	Tamil Nadu, Chennai
6	6/23/2018			Chennai	Tamil Nadu, Chennai
7	6/23/2018			Chennai	Tamil Nadu, Chennai
8	6/23/2018			Chennai	Tamil Nadu, Chennai
9	6/23/2018			Chennai	Tamil Nadu, Chennai
10	6/23/2018			Chennai	Tamil Nadu, Chennai
11	2/9/2019			Chennai	Tamil Nadu, Chennai
12	2/9/2019			Chennai	Tamil Nadu, Chennai
13	2/9/2019			Chennai	Tamil Nadu, Chennai
14	2/9/2019			Chennai	Tamil Nadu, Chennai
15	2/14/2019			Chennai	Tamil Nadu, Chennai
16	2/14/2019			Chennai	Tamil Nadu, Chennai
17	3/22/2019			Chennai	Tamil Nadu, Chennai
18	3/22/2019			Chennai	Tamil Nadu, Chennai
19	3/22/2019			Chennai	Tamil Nadu, Chennai
20	3/22/2019			Chennai	Tamil Nadu, Chennai
21	3/22/2019			Chennai	Tamil Nadu, Chennai
22	3/22/2019			Chennai	Tamil Nadu, Chennai

Sort Ascending

Sort Descending

Clear Sort

Clear Filter

Remove Empty

Text Filters

Tamil Nadu

(Select All Search Results)

Tamil Nadu

OK Cancel

Query Settings

PROPERTIES

Name

Orders Data

APPLIED STEPS

Source

Change Type

Promoted Headers

Changed Type

Kept First Rows

Changed Type with Locale

Capitalized Each Word

Inserted Merged Column

Merged Queries

Expanded Order Details

Kept Duplicates

Removed Duplicates

Sorted Rows

Filtered Rows

14 COLUMNS, 22 ROWS Column profiling based on top 1000 rows

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Query Settings

Layout Data Preview Columns Parameters Advanced Query Dependencies

Queries [3]

Sales target

Order Details

Orders Data

Table.SelectRows(#"Sorted Rows", each ([State] = "Tamil Nadu"))

Order ID	Order Date	CustomerName	State	City	Location
1	4/8/2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
2	4/8/2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
3	4/8/2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
4	4/8/2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
5	6/23/2018	Amisha	Tamil Nadu	Chennai	Tamil Nadu, Chennai
6	6/23/2018	Amisha	Tamil Nadu	Chennai	Tamil Nadu, Chennai
7	6/23/2018	Amisha	Tamil Nadu	Chennai	Tamil Nadu, Chennai
8	6/23/2018	Amisha	Tamil Nadu	Chennai	Tamil Nadu, Chennai
9	6/23/2018	Amisha	Tamil Nadu	Chennai	Tamil Nadu, Chennai
10	6/23/2018	Amisha	Tamil Nadu	Chennai	Tamil Nadu, Chennai
11	2/9/2019	Kalyani	Tamil Nadu	Chennai	Tamil Nadu, Chennai
12	2/9/2019	Kalyani	Tamil Nadu	Chennai	Tamil Nadu, Chennai
13	2/9/2019	Kalyani	Tamil Nadu	Chennai	Tamil Nadu, Chennai
14	2/9/2019	Kalyani	Tamil Nadu	Chennai	Tamil Nadu, Chennai
15	2/14/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
16	2/14/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
17	3/22/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
18	3/22/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
19	3/22/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
20	3/22/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
21	3/22/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai
22	3/22/2019	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai

Query Settings

PROPERTIES

Name

Orders Data

APPLIED STEPS

Source

Change Type

Promoted Headers

Changed Type

Kept First Rows

Changed Type with Locale

Capitalized Each Word

Inserted Merged Column

Merged Queries

Expanded Order Details

Kept Duplicates

Removed Duplicates

Sorted Rows

Filtered Rows

14 COLUMNS, 22 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 3:31 PM

Grouping and Aggregating Data

- To duplicate 'Order Details' Table, right click on the 'Order Details' in the 'queries' pane and select 'Duplicate' in the drop down menu.

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Query Settings

Layout Data Preview Columns Parameters Advanced Query Dependencies

Queries [4]

- Sales target
- Order Details
- Orders Data
- Order Details (2)

Table.Distinct("#Kept Duplicates")

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1275	-1148	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankerchief
B-25601	80	-56	4	Electronics	Electronic G
B-25602	168	-111	2	Electronics	Phones
B-25602	424	-272	5	Electronics	Phones
B-25602	2617	1151	4	Electronics	Phones
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	1355	-60	5	Clothing	Trousers
B-25603	24	-30	1	Furniture	Chairs
B-25603	193	-166	3	Clothing	Saree
B-25603	180	5	3	Clothing	Trousers
B-25603	116	16	4	Clothing	Stole
B-25603	107	36	6	Clothing	Stole
B-25603	12	1	2	Clothing	Hankerchief
B-25603	38	18	1	Clothing	Kurti
B-25604	65	17	2	Clothing	T-shirt
B-25604	157	5	9	Clothing	Saree
B-25608	1364	-1864	5	Furniture	Tables
B-25608	476	0	3	Furniture	Chairs
B-25608	257	23	5	Clothing	Hankerchief
B-25608	856	385	6	Electronics	Printers

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 3:38 PM

Query Settings

PROPERTIES

Name: Order Details (2)

APPLIED STEPS

- Source
- Use First Row as Headers
- Change Type
- Added Custom
- Changed Type
- Added Conditional Column
- Kept Duplicates
- Removed Duplicates

- To calculate count of each Id, or to calculate the average profit by each category or to calculate total amount by sub category, we use 'Group by' option in 'Home' tab.
- Count of each ID,

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Order ID

New column name: Count

Operation: Count Rows

Column:

OK Cancel

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [4]

- Sales target
- Order Details
- Orders Data
- Count of each ID

Table: Count of each ID

Order ID	Count
1 B-25601	4
2 B-25602	5
3 B-25603	8
4 B-25604	2
5 B-25608	4
6 B-25609	2
7 B-25610	6
8 B-25614	2
9 B-25616	4
10 B-25618	2
11 B-25621	3
12 B-25623	4
13 B-25625	3
14 B-25626	2
15 B-25628	3
16 B-25630	6
17 B-25633	3
18 B-25635	3
19 B-25638	4
20 B-25640	3
21 B-25643	5
22 B-25645	4
23 B-25648	4

Query Settings

PROPERTIES

Name: Count of each ID

APPLIED STEPS

- Source
- Use First Row as Headers
- Change Type
- Added Custom
- Changed Type
- Added Conditional Column
- Kept Duplicates
- Removed Duplicates
- Grouped Rows

2 COLUMNS, 265 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 4:12 PM

- Average Profit of Each Category,

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Category

New column name: Average Profit

Operation: Average

Column: Profit

OK Cancel

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns Keep Remove Rows Reduce Rows Sort Split Column Group By Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [5]

- Sales target
- Order Details
- Orders Data
- Count of each ID
- Average profit of Each C...

fx = Table.Group(#"Removed Duplicates", {"Category"}, {{ "Average Profit", each List.Average([Profit]), type nullable } })

Category	Average Profit
1 Furniture	10.44897959
2 Clothing	10.4278607
3 Electronics	45.04528302

Query Settings

PROPERTIES

Name

Average profit of Each Category

All Properties

APPLIED STEPS

- Source
- Use First Row as Headers
- Change Type
- Added Custom
- Changed Type
- Added Conditional Column
- Kept Duplicates
- Removed Duplicates
- Grouped Rows

2 COLUMNS, 3 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 4:17 PM

- For total amount of Each Sub-category,

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Sub-Category

New column name: Total Amount

Operation: Sum

Column: Amount

OK Cancel

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns Keep Remove Rows Reduce Rows Sort Split Column Group By Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [6]

- Sales target
- Order Details
- Orders Data
- Count of each ID
- Average profit of Each C...
- Total amount of each su...

Table.Group(*Removed Duplicates*, {"Sub-Category"}, {"Total Amount", each List.Sum([Amount]), type nullable

Sub-Category	Total Amount
1 Bookcases	47609
2 Stole	15081
3 Hankerchief	12856
4 Electronic Games	35084
5 Phones	38872
6 Saree	43675
7 Trousers	25596
8 Chairs	31129
9 Kurti	2843
10 T-shirt	5535
11 Tables	20011
12 Printers	51020
13 Accessories	18202
14 Furnishings	11676
15 Leggings	1749
16 Skirt	1534
17 Shirt	6149

Query Settings

PROPERTIES

Name

Total amount of each sub-category

APPLIED STEPS

- Source
- Use First Row as Headers
- Change Type
- Added Custom
- Changed Type
- Added Conditional Column
- Kept Duplicates
- Removed Duplicates
- Grouped Rows

2 COLUMNS, 17 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 4:18 PM

- To aggregate 'Sales Target' by total target amount by month of order date, use 'group by' on 'Home' Tab.

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Month of Order Date

New column name: Total Target Amount

Operation: Sum

Column: Target

OK Cancel

PowerBI Assignment 1 - Data Transformation and Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns Keep Remove Rows Reduce Rows Sort Split Column Group By Use First Row as Headers Replace Values Data Type: Decimal Number Merge Queries Append Queries Combine Files Combine

Queries [7]

- Sales target
- Order Details
- Orders Data
- Count of each ID
- Average profit of Each C...
- Total amount of each su...
- Total target amount by...

Month of Order Date 1.2 Total Target Amount

1	4/18/2026	31400
2	5/18/2026	31500
3	6/18/2026	31600
4	7/18/2026	33800
5	8/18/2026	33900
6	9/18/2026	34000
7	10/18/2026	36100
8	11/18/2026	36300
9	12/18/2026	36400
10	1/19/2026	43500
11	2/19/2026	43600
12	3/19/2026	43800

2 COLUMNS, 12 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 4:23 PM

Query Settings

PROPERTIES

Name

Total target amount by month of order b

All Properties

APPLIED STEPS

- Source
- Use First Row as Headers
- Change Type
- Changed Type
- Kept Duplicates
- Grouped Rows

Data Modeling

- To establish a relationship between two tables, select 'Manage Relationship' in 'home' Tab or any other manage relationship option, and establish the relationship.
- Tables are connected based on specifications and kept active

← Edit relationship



Select tables and columns that are related.

From table

Order Details

Amount	Category	Order ID	Profit	Profit Margin	Profit Status	Quantity
12	Clothing	B-25603	1	0.08333333333...	Profit	2
257	Clothing	B-25608	23	0.0894941634...	Profit	5
42	Clothing	B-25616	12	0.2857142857...	Profit	5

To table

Sales target

Category	Month of Ord...	Target
Furniture	Saturday, Apr...	10400
Clothing	Saturday, Apr...	12000
Furniture	Monday, May...	10500

Cardinality

Many to many (*:*)

Cross-filter direction

Both

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

← Edit relationship



Select tables and columns that are related.

From table

Order Details



Amount	Category	Order ID	Profit	Profit Margin	Profit Status	Quantity
12	Clothing	B-25603	1	0.08333333333...	Profit	2
257	Clothing	B-25608	23	0.0894941634...	Profit	5
42	Clothing	B-25616	12	0.2857142857...	Profit	5

To table

List of Orders



City	CustomerName	Order Date	Order ID	State
Indore	Hitika	Monday, Apri...	B-25627	Madhya Prad...
Indore	Ashmi	Thursday, Apr...	B-25637	Madhya Prad...
Indore	Yaanvi	Tuesday, May...	B-25645	Madhya Prad...

Cardinality

Many to one (*:1)



Cross-filter direction

Both



☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel